

A discrete, random, equilibrium vacuum: what it reproduces, what it predicts, what it cannot yet do

M. Cesaroni

Working paper, version 1 — October 2026. Code and results: github.com/m-cesaroni/hyperuniform-lattice-model. Documents under CC BY 4.0; code under MIT.

Abstract. Space is modelled as a three-dimensional *random* lattice of cells in mechanical equilibrium; light, matter and gravity are ways of using the faces, edges and vertices of the Voronoi tessellation. On that lattice the electromagnetic field (discrete exterior calculus) and a Kähler–Dirac field come out right without tuning: exact c , isotropy, no birefringence, no long-wavelength scattering, the relativistic energy–momentum relation as an algebraic identity. Time is not a coordinate of the model: it is the count of chirality inversions of a massive vibration, and time dilation, special and gravitational, is bookkeeping. From the same structure come the factor 2 in light deflection, Mercury’s perihelion from the saturation of a probabilistic occupation, and three generations from the geometry of channels; the model predicts a constant dark energy and a black-hole shadow 4.6 % larger than in general relativity, and dies if the data say otherwise.

Keywords: discrete spacetime · hyperuniform random lattice · discrete exterior calculus · Kähler–Dirac fermions · hydrogen atom on a lattice · emergent time · gravitational waves

1. The model, in standard terms

Three choices, all different from the known discrete-spacetime programmes (causal sets, dynamical triangulations, spin networks, cellular automata):

1. **Random lattice, not crystalline.** Points with a minimum distance, fixed density, periodic boundaries. No preferred axes; rotational invariance is statistical.
2. **Mechanical equilibrium.** The points repel each other with a finite-range potential and are relaxed until the forces vanish. The resulting lattice is *hyperuniform* ($S(k) \rightarrow 0$ as $k \rightarrow 0$, fluctuations $\propto$ surface) and locally isotropic in the directions of its neighbours. This condition is what makes everything else work.
3. **The Delaunay–Voronoi duality as the home of the fields.** 0-forms on cells, 1-forms on links (dual to Voronoi faces), 2-forms on triangles (dual to edges), 3-forms on tetrahedra (dual to vertices). Diagonal Hodge stars are ratios of dual measures. The EM field lives on the 1- and 2-forms; the spinor on all four (Kähler–Dirac).

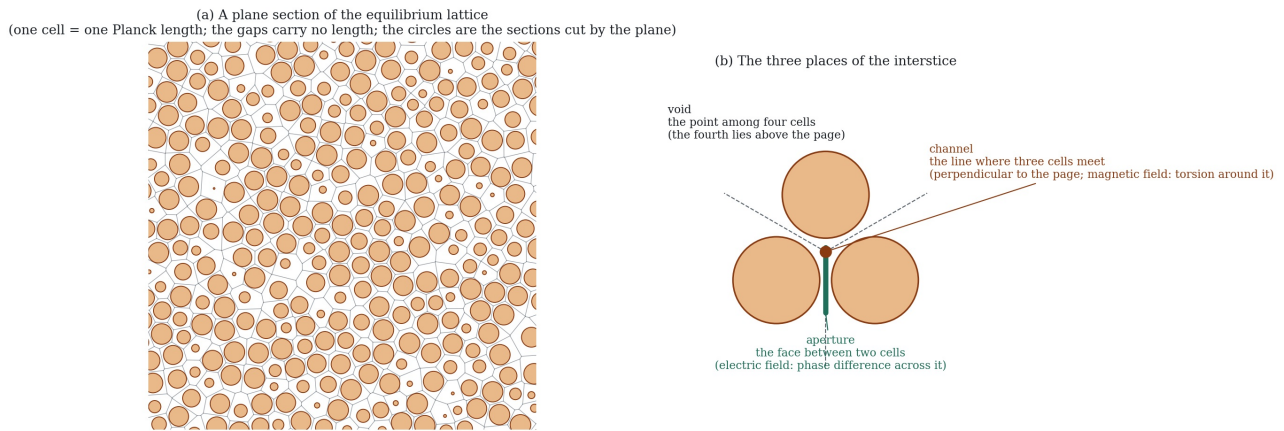


Figure 1. (a) A plane section of a 20 000-cell equilibrium lattice. One cell is one Planck length, and distance is the count of cells: the gaps between them are the interstice, which carries no length. In the mechanical model the cells are spheres of radius 0.44 of the mean spacing (packing fraction 0.70), a ratio that fixes the pushes and the equilibrium, not a measure of distance. The plane cuts each sphere at a different height, so the circles drawn are the sections of the cells, not their full size — a sphere cut near its centre appears large, one cut near its edge appears small. The lines are the Voronoi partition of the section. (b) The three places of the interstice: the aperture between two cells, the channel where three meet, the void among four.

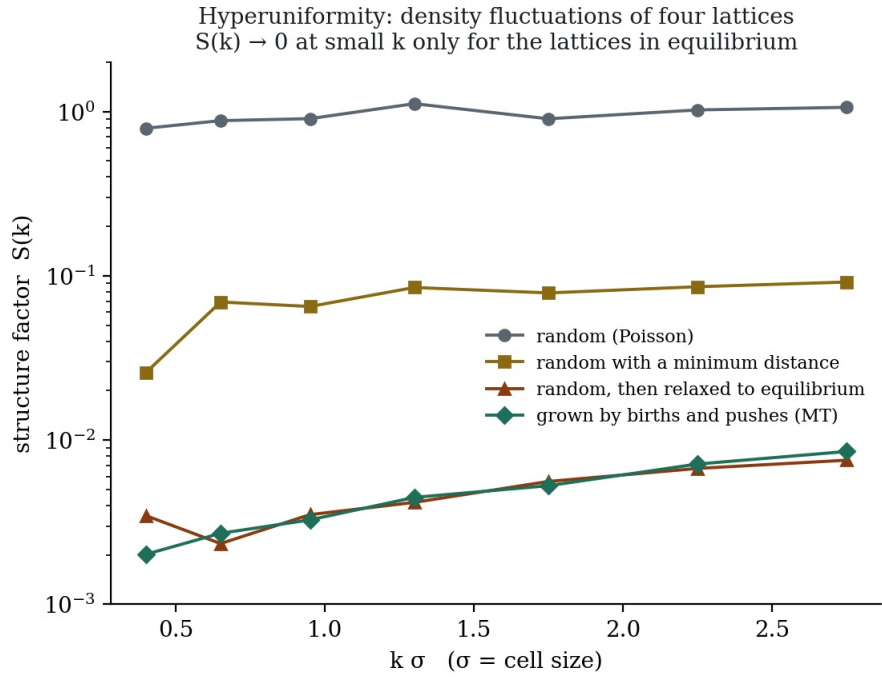


Figure 2. Structure factor $S(k)$ of four lattices: Poisson, random with a minimum distance, random then relaxed to equilibrium, grown by births and pushes. Only the two lattices in equilibrium are hyperuniform; their origin does not matter.

Time evolution: Hermitian Hamiltonian $K = d + d^\dagger$ (symmetrised with the Hodge stars); for light, an energy-exact leapfrog. Units: mean lattice step = 1, $c = 1$ in the continuum.

Matrix mechanics. The state is a vector of amplitudes on the elements of the complex; the position X_a is diagonal (coordinate of the dual centre of each element: cell, link midpoint, triangle centre, tetrahedron centre); K is sparse, with entries only between adjacent elements. For any diagonal matrix $([X_a, K])_{\alpha\beta} = (x_\alpha - x_\beta)_a K_{\alpha\beta}$, real and antisymmetric, hence $[X_a, K] = iV_a$ with V_a Hermitian: Heisenberg's equation $dX_a/dt = i[K, X_a] = V_a$ identifies V_a with the velocity (displacement along the passage $\times$ stiffness of the passage). Robertson's inequality follows: $\Delta X_a \cdot \Delta K \geq \frac{1}{2} |V_a|$. In the Schrödinger limit $[X_a, P_a] = i$ in lattice units (cell $\times$ radian per cell): $\hbar$ is a unit conversion. The uncertainty principle is verified from the spectrum of K (table), saturating the Gaussian minimum above one cell and reduced below it (minimal length).

2. Time

There is no time coordinate in the model. The lattice updates in steps — *batches* in the Italian documents — but a step is not an instant: it is a global counter that no observer can read. Physical time is something else, and it is defined operationally.

- **Proper time is a count.** A massive vibration inverts its chirality at frequency $2E/\hbar$ (Zitterbewegung); the mass operator is what inverts. The proper time of a particle is the number of inversions it has completed. On the lattice, for every eigenstate, $\langle \Gamma \rangle = m/E$, and the number of inversions per step is the share T of the budget, with $I^2 + T^2 = 1$ exact. $T = d\tau/dt = 1/\gamma$ follows without postulating relativity; at discrete steps the relation becomes $\cos \omega = \cos \mu \cdot \cos \kappa$ and tends to the continuous one for small angles.
- **Light has no time.** The EM field lives on the 1- and 2-forms and has no inversion term: no count, $\tau = 0$, consistent with relativity. This is also why light does not interact with itself and is not localisable: nothing happens to it between emission and absorption.
- **Gravitational dilation is a rhythm.** Near a mass the occupation n slows the inversions: $T^2 = e^{-n}$, whence redshift, GPS, Shapiro, and — with the same deficit read in space as well — $\gamma = 1$. The binding energy of hydrogen in Dirac's theory is exactly the deficit of the electron's proper time: $\langle \beta \rangle = \sqrt{1 - \alpha^2}$, $E = mT$.
- **Before the first vibration there is no time.** A lattice without massive excitations has steps but no clocks. The initial growth epoch has a number of steps, not a duration: "how long did inflation last" is not a well-posed question in the model, and a birth rate per step in that epoch is not a physical quantity. Time begins with the first excitation, i.e. with the first push on a closed shell of equilibrium cells.
- **The twin paradox and the energy–time uncertainty** are counts. The travelling twin spent more budget in motion and less in inversions. And Heisenberg's $\Delta E \cdot \Delta t \geq \hbar/2$ reads, in the model's own quantities, $\Delta \omega \cdot \Delta N \geq \frac{1}{2}$: ω is the phase advance per step of a state, N the number of steps counted, and to know how fast a quantity changes per step one must count enough steps. The inequality follows from the spectrum of the lattice operator K and is verified numerically ($\Delta \omega \cdot N_{1/2} = 0.80\text{--}0.87$ against the bound $\pi/4 = 0.785$; table in section 4). $\hbar$ enters only in the conversion to ordinary units.

Verified constraint: the conversion between steps and the proper time of a cosmic observer does *not* drift with the epoch — a fit to the DESI DR2 BAO distances excludes a clock factor depending on Ω_m (parameter $\approx 0 \pm 0.1$). The model's time does not rewrite cosmological chronology. Experimental reference for the time–mass identification: Lan et al., *Science* 339 (2013), the Compton clock.

3. Gravity and gravitational waves

The static field. A massive vibration pauses, and in pausing it returns repeatedly to the same cells. A cell visited in this way is *engaged*: it spends a share of its own steps for the engagement, and withdraws that share from whatever passes through it. The engagement spreads to the neighbours by contagion without loss at the source, so the field persists as long as the mass does. A critical contagion gives a mean number of engagements per cell $n \propto 1/r$ (measured exponent -0.986); the occupation of a cell — the probability that it carries at least one engagement — is $1 - e^{-n}$, and the rate of time is $T^2 = e^{-n}$. The saturation of this probability is what fixes Mercury's perihelion advance ($42.99''$ per century) without an added term, and what removes the horizon: T^2 never vanishes. The same deficit read in space as well as in time gives the factor 2 of light deflection. What is not derived is the strength: G corresponds to about 5–6 engagements per inversion with the cell at the Planck scale, close to half the number of stiff neighbours ($N/2 = 5.85$), a hypothesis with a mechanism — an inversion engages the half-sphere of neighbours towards which it returns — but not yet a calculation. profile derived strength: hypothesis

The carrier and its speed. Gravity lives in the interstice, and a purely interstitial propagation would be timeless, hence instantaneous; it is not, so the propagation must involve the cells. The carrier is the elastic push: an engagement alters the pushes of the neighbouring cells, and altered pushes are transmitted one cell per step — the only transfer rule the lattice has, the same that gives light its speed. The random walk used to obtain the $1/r$ profile survives only as the statistics of engagement: the static profile is the same because both solve the discrete Poisson equation, while the transient travels at c . This is the sense in which the equality of the speeds of light and of gravity, measured by GW170817 to one part in 10^{15} , is not a coincidence in the model but a consequence of there being one rule of transfer. mechanism identified; lattice verification pending

What a gravitational wave is. The static field and the wave are not the same thing. The field is the fraction of steps spent by engaged cells, with the contraction of the interstice around a mass; the wave is the directional, time-dependent part of the same elastic push. It acts along a direction of the interstice and moves cells: since nothing below one cell has meaning for a cell, the displacement is a whole cell with a probability. Dynamically, the wave redistributes *adjacencies*: cells enter the chains along the stretched axis and leave the chains along the compressed one, at constant number of cells. Free mirrors stay on their cells and light counts more or fewer cells between them — this is what an interferometer measures — while a bound rod re-forms its own count and measures nothing.

Its shape. Three modes are conceivable and two are excluded by structure. No breathing mode: outside a spherical source the contraction depends only on the enclosed occupation (the discrete form of Gauss's law and of Birkhoff's theorem), so a pulsating sphere emits nothing — the exclusion does not come from any incompressibility of the lattice, which does contract. No displacement mode: the pushes close on every cell, which is the conservation of momentum. What remains is the cross — stretching along one axis, shortening along the perpendicular one, at constant cell count, hence traceless — in its two orientations: two polarisations, as for light, because the interstice has two directional properties, the stretching of the apertures and the torsion of the channels, which also carry the wave. reason established

Its amplitude and its source. Two contracted regions in rotation emit the cross at twice the orbital frequency, with a static scalar part outside. For the first observed merger (GW150914) the strain of the model, $h = 2 n_\mu (v/c)^2$, where n_μ is the occupation at the detector due to the source's mass and v the orbital speed, gives $2.1 \cdot 10^{-21}$ — the same number as the quadrupole formula, because the two expressions coincide in the weak-field limit. How the wave is launched at the moment of contact of the two bodies — whether its direction is fixed by a rebound at the point of contact — is a hypothesis under study, not a result. amplitude formation at merger: open

What must be verified on the lattice. That a source continuously engaging a half-sphere produces a static deformation $\propto 1/r$ and a transient arriving at distance r after r steps; that the shear wave propagates isotropically and without dispersion at long wavelengths; that two rotating contracted regions emit it at twice their frequency; and that its speed coincides with that of light to the precision GW170817 requires — a new constraint, because light travels with the geometric stiffnesses of the apertures and the wave with the elastic pushes between cells. simulation to be written

4. Numerical results (lattices from $4 \cdot 10^3$ to $2.7 \cdot 10^6$ cells)

Quantity	Result	Reference	Status
Speed of light (long waves)	1.0002 ± 0.0005	1	no normalisation
Anisotropy of c between directions	1.3 % (small lattice)	0	decreases with size
Dispersion: $c(k) = c_0[1 - \beta(ka)^n]$	$n = 1.84 \pm 0.06$, $\beta \approx 0.01\text{--}0.14$	linear excluded (Fermi)	quadratic
Velocity difference between helicities	$< 10^{-3}$ (spin $\frac{1}{2}$), $< 10^{-7}$ (EM)	$< 10^{-16}$ (GRB)	compatible
Scattering of light, waves ≥ 16 steps	$< 1.2 \cdot 10^{-7}$ per unit time	sharp galaxies	not detected
Scattering law at onset	$\Gamma \propto kp$, $p \approx 6\text{--}8$ (eq.), 2 (spin in cells)	$p \geq 3.4$ suffices at TeV	the k^2 was doubling
Scattering ratio equilibrium / non-equilibrium lattice	$1/10 - 1/100$	—	equilibrium matters
Inversion share $T = \langle \Gamma \rangle$ vs m/E , eigenstate by eigenstate	deviation $7 \cdot 10^{-15}$	$1/\gamma$	time = count
Oscillation along the spin axis (packet at rest)	period π/E , transverse $< 10^{-2}$	Zitterbewegung	measured
Anticommutation $\{K, \Gamma\}$ and $E^2 = K^2 + m^2$ (Kähler–Dirac)	0 and 10^{-14} on all eigenstates	identity	theorem
Discrete-step version	$\cos \omega = \cos \mu \cdot \cos k$, deviation 10^{-15}	spherical Pythagoras	theorem
Spurious low-energy modes (doubling)	258/260 (spin in cells); 0 (Kähler–Dirac)	Nielsen–Ninomiya	2 residual tastes
Field profile from a critical contagion	$r^{-0.986}$	$1/r$	discrete Poisson
Occupation from independent engagements	$1 - e^{-n}$, var/mean = 1.000	—	Poisson
Hyperuniformity: $S(k)$ at small k	0.002 (eq.) vs 0.05 (RSA) vs 0.9 (Poisson)	—	measured
$\Delta x \cdot \Delta k$ from $\langle K^2 \rangle$ for Gaussian packets of width s	0.49 ($s = 1\text{--}2$ cells); 0.45 ($s = 0.6$)	$\frac{1}{2}$	spectrum of K
$\Delta \omega \cdot N^{1/2}$ (survival under e^{iK})	0.80 – 0.87	$\geq \pi/4 = 0.785$	Mandelstam–Tamm

Hydrogen, Schrödinger of the model on $2.7 \cdot 10^6$ cells ($a_0 = 4$ cells, box $32 a_0$)	1s -1.2% ; 2p $+0.17\%$; 2s -0.28%	$-\mu\alpha^2/2n^2$	Laplacian from Hodge weights
2s–2p degeneracy; 2p triplet; 3d under a spherical wall	0.45 %; $4 \cdot 10^{-5}$; $1 \cdot 10^{-4}$	0; 0; 0	no symmetry imposed
Lyman- α oscillator strength ((2p X 1s)); extrapolation to infinite resolution and box (6 runs)	0.425; 0.410 – 0.414	0.4162	+2.2 % / -1 %
n = 3 manifold (box $50 a_0$): 3s, 3p, 3d degenerate among themselves and with Rydberg/9; sums $1s \rightarrow n=3$, $2s \rightarrow n=3$, $2p \rightarrow n=3$	0.4 %; 0.082 / 0.468 / 0.699	0; 0.0791 / 0.4349 / 0.7094	+3.9 % / +7.7 % / -1.5 %
Balmer 2p $\rightarrow$ 3d oscillator strength (boxes $42 a_0$ and $32 a_0$)	0.678; 0.690	0.6958	-2.6 % / -0.8 %
Controls: packing fraction 0.60–0.80; periodic vs spherical wall (n = 2); 10 % weight noise; $10\times$ worse relaxation	$\Delta f \leq 1\%$; identical to 0.05 %; $\Delta E \leq 0.4\%$; $\Delta E \leq 0.3\%$	—	stable
Apertures around channels (Delaunay triangles)	typical imbalance 0.21–0.23 rad at $\phi = 0.70$	Koide angle 0.222	depends on ϕ

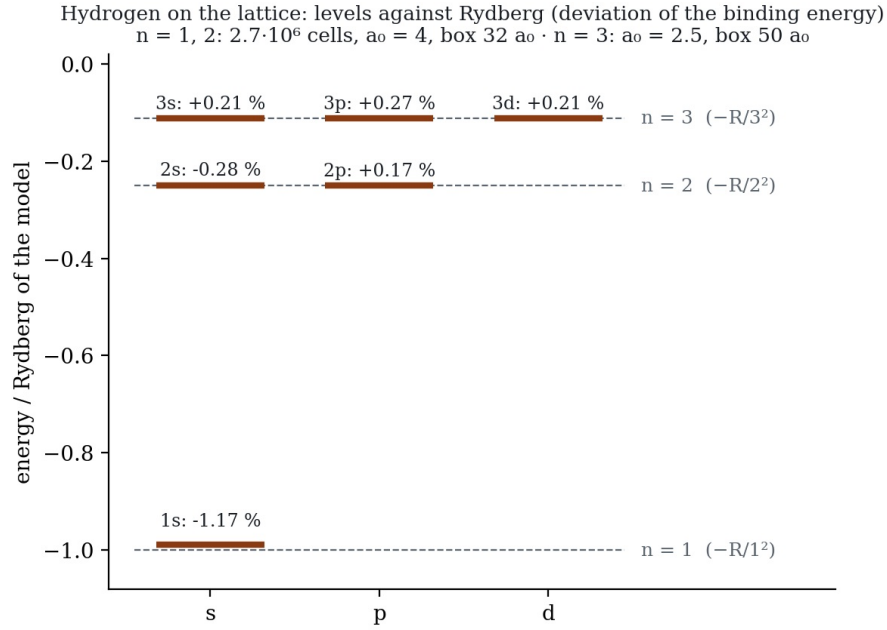


Figure 3. Hydrogen on the lattice: levels against Rydberg with the deviation of the binding energy. n = 1, 2 from the 2.7-million-cell run ($a_0 = 4$ cells, box $32 a_0$); n = 3 from the $50 a_0$ box ($a_0 = 2.5$).

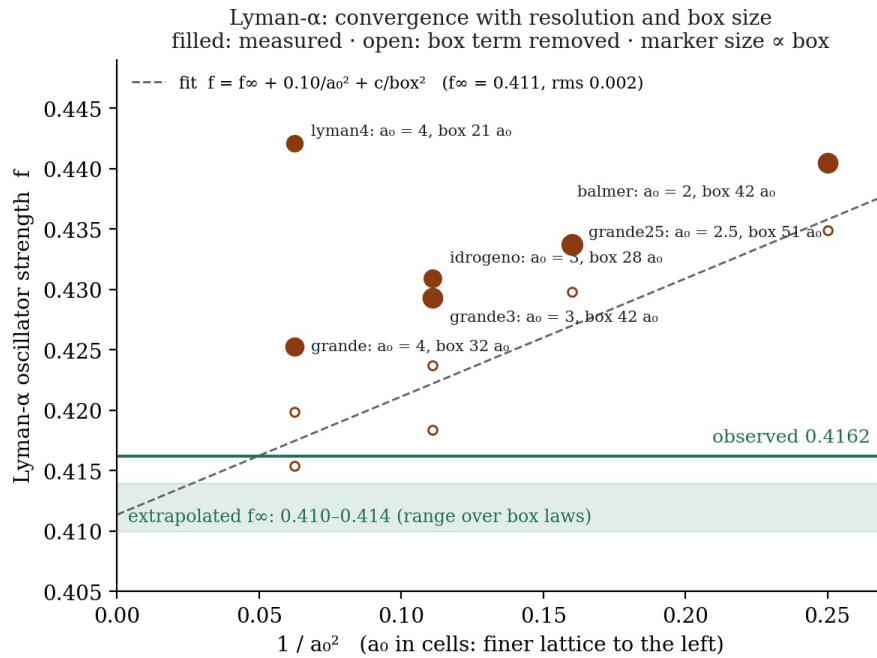


Figure 4. Lyman- α oscillator strength of six runs against $1/a_0^2$, with the box term removed (open markers), the fit $f = f_\infty + 0.10/a_0^2 + c/\text{box}^2$, the observed value 0.4162 and the extrapolated range.

5. What it reproduces, and with how many assumptions

With no free parameter

- Relativistic energy–momentum relation and the budget $v^2 + (1/y)^2 = 1$ as consequences of the anticommutation motion–inversion; proper time as the count of inversions.
- The hydrogen spectrum in the Rydberg scale, the accidental degeneracies and the intensities of Lyman- α and Balmer $2p \rightarrow 3d$ from the lattice Schrödinger equation, without tuned constants; the cubic splitting of the 3d states belongs to the periodic images and disappears under a spherical wall.
- Zitterbewegung frequency $2E/\hbar$, oscillation along the spin axis, amplitude $c/2m$.
- Deflection $1.7516''$, Shapiro $123.7 \mu s$, GPS $+38.5 \mu s/day$ from the principle "one deficit read in time and in space" ($\gamma = 1$).
- Mercury $42.99''/century$ from the saturation of the occupation ($\beta = 1$), with no added term.
- Three generations, no fourth: three faces around a Voronoi edge; Koide's relation $Q = 2/3$ as an identity of the three projections at 120° .
- No magnetic monopoles ($d^2 = 0$) and no light–light interaction (linearity), by structure.

With one parameter or one identification

- Hierarchy $1 : 207 : 3477$ if the dipole angle is the typical imbalance of the channels, which comes out $0.21\text{--}0.23$ rad at packing fraction 0.70 (0.28 at 0.60 ; $0.17\text{--}0.21$ at 0.80).
- Neutrino masses $0.38 / 8.9 / 50.7$ meV (Brannen's scheme): two parameters, reproduces both Δm^2 .
- G equivalent to $5\text{--}6$ "engagements per inversion" with the cell at the Planck scale; $\approx$ half of the 11.7 neighbours that fix $\alpha = 1/N^2$.
- Uniform chirality of the universe from the first birth-ordered tetrahedron (sign); the amount of the baryon asymmetry remains to be derived.

6. The known obstacles, by name

- **Lorentz invariance.** A preferred frame exists (the lattice). It is statistically invisible: corrections are quadratic, $E_{QG,2} \approx 2.7 E_{\text{Planck}}$ against current limits $\approx 10^{-8} E_{\text{Planck}}$; the time of flight of GRB 090510 comes out $2 \cdot 10^{-19}$ s.
- **Fermion doubling.** A spinor placed in the cells produces a constant density of spurious modes ($p = 2$ in the scattering: conversion into lattice modes). Kähler–Dirac removes them at the price of two nearly degenerate copies — tastes, in lattice terms — ($0.1\text{--}0.3\%$ splitting at $\lambda = 15$ steps). Not solved: the main open problem on matter.
- **Aether.** It is Lorentz's aether made invisible by hyperuniformity: real parts, undetectable by waves longer than the cells. Where it differs from GR it is measurable (below).
- **Speed of the gravitational contagion.** The $1/r$ profile was first obtained with a random walk, which as a propagation mechanism would be incompatible with GW170817 by thirty orders of magnitude. The mechanism has since been identified — the elastic push, transmitted one cell per step, with the random walk surviving only as the statistics of engagement (section 3) — and the profile is the static solution of the same equation. What remains is the verification on the lattice: the transient at c and the two-polarisation wave have been argued, not yet simulated.
- **Vacuum energy and masses.** The model does not solve the hierarchy: it translates it into "why inversion is rare" (m/m_p per step). The 120 orders of magnitude of the vacuum energy remain.

7. Falsifiable predictions

Prediction	Value	Test	If false
Constant dark energy ($w = -1$)	$\Lambda; p_0 = 3H_0 t_P \approx 4 \cdot 10^{-61}$ per cell per step	final DESI, Euclid (2027–2030)	a $w_0 w_a$ at 5σ requires a relaxation dynamics not yet written
Black-hole shadow	$+4.6\%$ relative to GR ($T^2 = e^{-n}$, no horizon)	ngEHT, Sgr A* (uncertainty today $\approx 9\%$)	the strong-field saturation is wrong
Second-order solar deflection	$+0.69 \mu as$ relative to GR	LATOR-type missions	same
No fourth generation	exactly 3	already verified (Z, Higgs)	—
Sum of neutrino masses	60 meV, normal ordering, Majorana	cosmology 2030–34	the three-projection neutrino scheme falls
No vacuum birefringence; dispersion only quadratic	—	GRB polarisation, Fermi/CTA	the lattice would have a geometric handedness

8. What it cannot do (today)

1. Separate or interpret the two copies of the Kähler–Dirac vibration.
2. Construct the "cell–void exchange" mass operator that anticommutes with K (the naive version transports instead of inverting).
3. Fix the packing fraction of the vacuum: Koide's angle is a one-parameter coincidence until ϕ is derived.
4. The growth of structure with this gravity (needed to compare the 60 meV with lensing limits).
5. The weak sector (W, Z, coupling to one chirality) and the amount of the baryon asymmetry.
6. The lattice dynamics of gravitational waves: the simulation of the shear wave, its speed against that of light, and its emission by a rotating

source.

9. Reproducibility

All numbers in the table come from Python scripts (numpy, scipy) that run on a laptop: lattice generation (RSA, equilibrium, growth by adjacent births, CCC crystal as control), measurements of $S(k)$ and number variance, light propagation with discrete exterior calculus (leapfrog, energy conserved to 10^{-12}), Weyl and Kähler–Dirac operators with exact diagonalisation on small lattices, Monte Carlo contagion. Times: 30 minutes for $3 \cdot 10^4$ cells, an hour and a half for $1.3 \cdot 10^5$. The scripts are in the `code/` folder of the source repository (github.com/m-cesaroni/hyperuniform-lattice-model; archived releases at [doi:10.5281/zenodo.23189702](https://doi.org/10.5281/zenodo.23189702)) (`mt_trasparenza.py`, `mt_luce.py`, `kdlib.py`, `mt_idrogeno_pc.py`), and in `code/analysis/` those that produce the individual numbers of the table (channel geometry, neighbours, uncertainty, commutator, sectors, pushes, DESI fit). The gravity numbers — $1/r$ profile from the contagion, Mercury from saturation — come from scripts of an earlier phase not yet consolidated in the repository: they will be added in a later version.

10. Literature it rests on

1. Desbrun, Hirani, Leok, Marsden — Discrete exterior calculus (2005); Bossavit — Computational electromagnetism.
2. Nielsen, Ninomiya — No-go theorem for lattice chiral fermions (1981); Christ, Friedberg, Lee — Random lattice field theory (1982); Rabin — Kähler–Dirac fermions on lattices (1982).
3. Torquato, Stillinger — Hyperuniformity (2003); Leseur, Pierrat, Carminati — Transparency of hyperuniform media (2016); Donev, Stillinger, Torquato — Jammed packings are hyperuniform (2005).
4. Feynman, Hibbs — Checkerboard (1965); Kac (1974); Gaveau, Jacobson, Kac, Schulman — Telegraph/Dirac and Brownian motion (1984); Nelson — Stochastic mechanics (1966).
5. Koide (1982); Brannen — Koide for neutrinos (2006).
6. Sorkin — Causal sets; Ambjørn, Jurkiewicz, Loll — CDT; 't Hooft — Cellular automaton interpretation; Einstein — Äther und Relativitätstheorie (Leiden, 1920).
7. DESI DR2 (2025–26); Vasileiou et al. — Fermi constraints on Lorentz violation (2013); EHT Sgr A* (2022); Will — PPN confrontation (2014).

Appendix A. Glossary

The vocabulary of the model, each term with its Italian original, its meaning in the model and its nearest counterpart in standard physics. Where the correspondence is only partial, the third column names the closest concept, not an identity.

Term	In the model	Standard counterpart
THE LATTICE		
Cell <i>cella</i>	The elementary unit of space: spherical, never in contact with its neighbours, one Planck length across; distance is the count of cells, and the gaps between them carry no length. Cells carry states and time; nothing physical is smaller than a cell.	Lattice site; Delaunay vertex; Voronoi cell. A Planck-scale quantum of space.
Step <i>batch</i>	The global update counter of the lattice. One step is not an instant: it is one application of the rules, converted to one Planck time only when translated into seconds. Steps carry time only where a massive vibration counts them.	Discrete update step of a lattice dynamics. Not a time coordinate.
Interstice <i>interstizio</i>	The space between cells. It carries the conditions of passage — phase differences, torsions, contraction, agreement — but no states and no time. Elastic, with stiffnesses fixed by geometry.	The Voronoi/Delaunay complex (faces, edges, vertices) with its Hodge weights; the elastic medium.
Aperture <i>varco</i>	The passage between two neighbouring cells. Seat of the electric field (a phase difference across it). Stretching stiffness = face area / length.	Voronoi face ↔ Delaunay link; a 1-form; Hodge star $\star_1$.
Channel <i>canale</i>	The line onto which three cells open. Seat of the magnetic field (the torsion left by differences going round it). Three faces around a channel → three generations. Torsion stiffness = channel length / triangle area.	Voronoi edge ↔ Delaunay triangle; a 2-form; Hodge star $\star_2$.
Void <i>vuoto</i>	The region among four cells, the fourth part of the interstice. Light never enters it; the spin vibration does. The cell/void agreement is chirality.	Voronoi vertex ↔ Delaunay tetrahedron; a 3-form.
Connection <i>collegamento</i>	The segment joining the centres of two neighbouring cells; the dual reading of an aperture. In units of the mean spacing: typical length 0.97, short below 0.8, inert beyond 1.3 (ratios of the mechanical model, not physical distances: distance is the count of cells).	Delaunay edge.
Stiff neighbour <i>vicina rigida</i>	A neighbouring cell whose connection is shorter than 1.3 mean spacings and therefore transmits pushes rigidly. $N = 11.7$ per cell in the equilibrium vacuum; $\alpha = 1/N^2$.	Effective coordination number.
Push <i>spinta</i>	The elastic force a cell exerts on a neighbour through the interstice. Equilibrium is the closing of the pushes on every cell; the transmission of altered pushes, one cell per step, carries gravity.	Interparticle force; momentum conservation; the elastic field of the medium.
Equilibrium <i>equilibrio</i>	The state in which the pushes close on every cell: no phase difference, no cost, no time. Such a lattice is hyperuniform, isotropic and transparent. The condition on which every result depends.	Mechanically relaxed (jammed) random packing; hyperuniformity, $S(k) \rightarrow 0$.
Hodge weights <i>pesi di Hodge</i>	The ratios of dual measures (face/length, length/area) that weight every passage. They are the only "constants" of the interstice, and they are fixed by the positions of the cells.	Diagonal Hodge star of discrete exterior calculus.
EXCITATIONS AND THE BUDGET		
Vibration <i>vibrazione</i>	An excitation of the lattice: a pattern of amplitudes on cells and interstice that is re-excited cell after cell. A particle is a vibration; it does not move, it is re-ignited.	Field quantum; wave packet.
Ignition <i>accensione</i>	The re-excitation of the next cell along the direction of motion. Light is a sequence of ignitions, one cell per step, without ever entering a cell.	Propagation at c ; the photon has no proper time.
Budget <i>budget</i>	The unit total available to every vibration, partitioned over orthogonal axes: $I^2 + T^2 + qIG^2 = 1$. The relativistic mass-shell relation written in units of budget.	Normalisation of the four-velocity; $E^2 = p^2 + m^2$.
Propagation share I <i>quota di propagazione</i>	The share of the budget spent in not pausing. Light has $I = 1$ always.	v/c .
Time share T <i>quota di tempo</i>	The share spent in pauses: each pause is one tick of the clock. $T = dt/dt$.	$1/\gamma$; in Dirac's theory $\langle \beta \rangle = m/E$.
Field share qIG <i>quota ceduta al campo</i>	The share withdrawn by engaged cells; depends on position only.	Gravitational potential; redshift factor.
Pause <i>sosta, stazionamento</i>	The chirality inversion of a massive vibration: the part that passes through apertures and voids is reversed. The count of pauses is proper time. At rest, one inversion every $1/m$ steps.	Dirac mass term; Zitterbewegung; $\langle \Gamma \rangle = m/E$ on every eigenstate.
Inversion amplitude <i>ampiezza di inversione</i>	The rate of pauses per step: the mass of a vibration in lattice units. For the electron $4 \cdot 10^{-23}$ per step; for the Planck mass, 1.	Mass in Planck units, m/m_P .
Phase difference <i>scarto di fase</i>	A mismatch of phase across an aperture. It costs the square of the mismatch; equal differences add, opposite ones cancel. Written by a vibration, it is work spent; crossed, it is reflection; released, it is light.	Electric field (1-form); gauge-field energy density.
Torsion <i>torsione</i>	The circulation of phase differences around a channel.	Magnetic field (2-form); the curl.
Agreement <i>accordo</i>	The relation — in phase or in antiphase — between the vibration on the cells and the one in the voids. It distinguishes the two copies of the spin vibration and is what the weak interaction acts upon.	Chirality; the weak doublet; a Higgs-like reflective state of the interstice.
Orientation (handedness) <i>verso, mano</i>	The single chirality of the lattice, decided by chance at the first tetrahedron and inherited by adjacency. It fixes the handedness of the weak interaction and which phase is called matter.	Parity violation; the sign of the matter–antimatter asymmetry.
Kähler–Dirac vibration <i>vibrazione di Kähler–Dirac</i>	The spin vibration distributed over all four parts of the interstice: cells, apertures, channels, voids. Free of spurious modes; it comes in two nearly degenerate copies.	Kähler–Dirac (geometric) fermion; two tastes.
Copies <i>copie, doppioni</i>	The two nearly identical spin vibrations produced by the cell/void duality (splitting 0.1–0.3 %). Read as the weak doublet, not as a defect.	Fermion doubling (Nielsen–Ninomiya); taste multiplicity.

K <i>operatore del reticolo</i>	The sparse Hermitian operator that advances the state by one step; one entry per passage. $[X_a, K] = iV_a$ is an identity.	Lattice Hamiltonian $d + d^\dagger$ (symmetrised); the Kähler–Dirac operator.
GRAVITY		
Engagement <i>impegno</i>	What happens to a cell visited repeatedly by a pausing vibration: it spends part of its own steps for the engagement and withdraws a share from whatever passes through it. The mean number of engagements per cell, n , falls as $1/r$.	Source term of the gravitational field.
Occupation <i>occupazione</i>	The probability that a cell carries at least one engagement: $1 - e^{-(n)}$. It saturates; the rate of time is $T^2 = e^{-(n)}$. No horizon, no singularity.	Metric time dilation, g_{00} ; saturation replaces the Schwarzschild singularity.
Contagion <i>contagio</i>	The spreading of engagement to neighbouring cells without loss at the source. Critical contagion gives the $1/r$ profile; the transient travels at one cell per step.	Poisson's equation (static); retarded propagation of the field.
Contraction <i>contrazione</i>	The denser packing of cells near a mass, by the same fraction as the slowing of time: the same deficit read in space. A free rod counts more cells; a bound rod the same number.	Spatial part of the metric; the factor 2 in light deflection; what interferometers measure.
HADRONS AND THE COSMOS		
Vortex <i>vortice</i>	A closed circulation of internal phase around a channel: one full turn, one third per face. The proton. The number of turns is conserved.	Baryon; baryon number as winding number; confinement.
String (vortex line) <i>corda</i>	The line of cells each carrying the full winding; energy proportional to length; it pinches into two turns when stretched.	QCD flux tube; dual superconductor ("t Hooft–Mandelstam); pair creation, jets.
Scale Λ <i>rigidità del vuoto contro gli avvolgimenti, ϵ_0</i>	The stiffness of the equilibrium vacuum against windings: $10^{-80} E_P$ per engaged cell. Measured in four ways (string tension, proton mass, tube width, g_A), not derived.	$\sqrt{\sigma} = 440 \text{ MeV}$; Λ_{QCD} ; the constituent quark mass.
Reflection of the aperture <i>riflessione del varco</i>	The function that assigns a cost to a phase mismatch across an aperture. The electron amplitude (temporal mismatch) and the scale Λ (spatial mismatch) are two values of it; deriving the function is the central open problem.	Mass generation; the hierarchy problem in MT's terms.
Birth <i>nascita</i>	The creation of a new cell at the centre of a connection stretched beyond $1.3 \ell_P$. Almost everywhere at the beginning (inflation), nowhere at equilibrium, almost never today: $p_0 = 3H_0 t_P \approx 4 \cdot 10^{-61}$ per cell per step.	Cosmic expansion; the cosmological constant ($w = -1$).
Dark matter <i>materia oscura (mtdm)</i>	Frozen spherical contractions of the interstice: regions whose connections were never stretched. No events, no time, mass proportional to radius.	Cold, collisionless dark matter.
LABELS USED IN THE DOCUMENTS		
demonstrated <i>dimostrato</i>	A theorem, or a numerical result reproduced by the code in the repository.	
coherent <i>coerente</i>	A mechanism that reproduces the data but still depends on a choice, or on a scale, not yet derived.	
open <i>aperto</i>	A problem not yet solved, stated with its target number.	

Conventions. Lattice units: mean cell spacing = 1 (one Planck length), one step = one Planck time, $c = 1$. Translation choices fixed for all English documents: *pause* (not "stop" or "rest"), *aperture* (not "gap"), *channel*, *void*, *engagement* (not "occupation", which is reserved for the probability $1 - e^{-(n)}$), *agreement*, *orientation* or *handedness* for the Italian *verso/mano*, *birth*, *step* for *batch*.